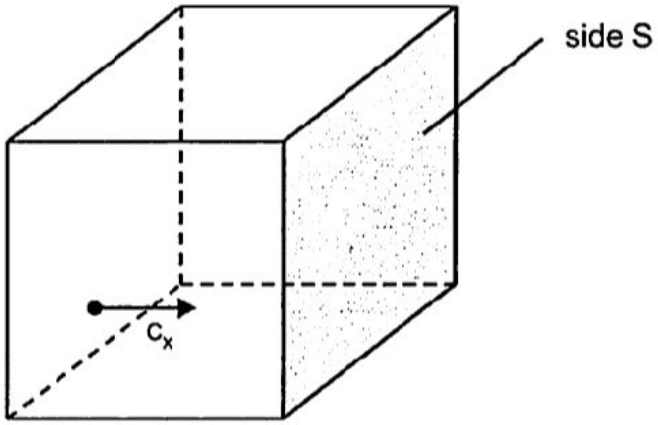
3	(a)	The kinetic theory of gases is based on some simplifying assumptions. For example, the molecules of the gas are assumed to behave as hard elastic identical spheres. State the basic assumption about the molecules of an ideal gas based on		
		(i)	the nature of their movement,	
			.....	
			.....	[1]
		L1	Accept either one: **Random motion** OR **Constant velocity between collisions**	**B1**
		(ii)	their volume.	
			.....	
			.....	
			.....	[2]
		L1	**Total volume of molecules is negligible** **compared to volume of containing vessel /container.**	**B1** **B1**
	(b)	A cube of volume V contains N molecules of an ideal gas. Each molecule has a component c_x of velocity normal to one side S of the cube, as illustrated in Figure 3.1.  **Figure 3.1** The pressure p of the gas due to the component c_x of velocity is given by the expression $$pV = Nm\langle c_x^2 \rangle$$ where m is the mass of a molecule and $\langle c_x^2 \rangle$ is the mean-square-speed of the molecules in the direction normal to side S of the cube in Figure 3.1. Show and explain how the above expression leads to the relation	

$$pV = \frac{1}{3}Nm\langle c^2\rangle$$

where $\langle c^2\rangle$ is the mean-square-speed of the molecules.

[3]

L3 *Explain how the kinetic theory of gases is applied in your answer.*

Based on the assumptions of the kinetic theory of gases, **B1**

1. **all gases consist of a very large number *N* of identical particles, and,**
2. **all the particles move randomly (at varying speeds and move in all directions with equal probability).**

These imply that the mean square speed component in any one of the three mutually perpendicular directions (*x*, *y* and *z*) is the same. **B1**

i.e. $\langle c_x^2\rangle = \langle c_y^2\rangle = \langle c_z^2\rangle$

$\langle c^2\rangle = \langle c_x^2\rangle + \langle c_y^2\rangle + \langle c_z^2\rangle$ **B1**

Therefore,

$\langle c_x^2\rangle = \frac{1}{3}\langle c^2\rangle$

Hence, $pV = Nm\langle c_x^2\rangle$ **leads to** $pV = Nm.\frac{1}{3}\langle c^2\rangle = \frac{1}{3}Nm\langle c^2\rangle$ A0

(c) Neon gas behaves as an ideal gas.

A mass of 20.0 g of neon gas occupies a volume of 2.24×10^4 cm^3 at a pressure of 1.01×10^5 Pa and a temperature of 273 K.

Deduce the root-mean-square (r.m.s.) speed of neon atoms at a temperature of

(i) 273 K,

r.m.s. speed = ……m s^{-1} [3]

L2 Note that **total mass** of gas = Nm = 20.0 g

$pV = \frac{1}{3}Nm\langle c^2\rangle$

$(1.01 \times 10^5)\,(2.24 \times 10^4 \times 10^{-6}) = \frac{1}{3}(20.0 \times 10^{-3})\,\langle c^2\rangle$ M1

$\langle c^2\rangle = 3.3936 \times 10^5$ C1

$\sqrt{\langle c^2\rangle} = \sqrt{3.3936\times10^5} = 582.55 = 583$ m s^{-1} A1

(ii) 27 °C.

r.m.s. speed = ……m s^{-1} [2]

L2 $\langle c^2\rangle \propto T$

$$\frac{\langle c^2\rangle_{27°C}}{\langle c^2\rangle_{273\ K}} = \frac{27 + 273.15}{273}$$

$$\frac{\langle c^2\rangle_{27°C}}{3.3936 \times 10^5} = \frac{27 + 273.15}{273}$$ M1

$\langle c^2\rangle_{27°C} = 3.7311 \times 10^5$

$\sqrt{\langle c^2\rangle_{27°C}} = 610.83 = 611$ m s^{-1} A1

[Total: 11]